

Nikhil Tammalla

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SUMMARY

Final-year Computer Science student with hands-on experience building AI-powered applications, LLM pipelines, and scalable backend systems. Published in IEEE across 3 projects. Proficient in Python, FastAPI, and RAG-based architectures.

EDUCATION

Amrita Vishwa Vidyapeetham — GPA: 7/10

B.Tech in Computer Science Engineering

Bangalore, KA

May 2022 – June 2026

SR College — GPA: 8.7/10

Maths, Physics, Chemistry

Warangal, TG

May 2020 – April 2022

SKILLS

Programming Languages: Python • Java • C • JavaScript

Backend & Systems: FastAPI • Django • Node.js • REST APIs • WebSockets • Socket.IO • Real-Time Systems • Modular Backend Design

Frontend Development: Responsive UI Design • HTML • CSS • JavaScript

Machine Learning & AI: scikit-learn • Supervised Learning • Classification • Regression • Feature Engineering • Model Evaluation • Computer Vision

Deep Learning & Computer Vision: CNNs • U-Net • Attention Mechanisms • Residual Networks • TensorFlow/Keras • Medical Image Segmentation

LLMs & Generative AI: LLM Integration • RAG • Prompt Engineering • LangChain • Semantic Search • AI Chatbots

Databases & Storage: MySQL • SQLite • Supabase • Qdrant

Tools & Platforms: Git • VS Code • Linux • Postman

Soft Skills: Problem Solving • Communication • Analytical Thinking • Fast Learner • Collaboration • Adaptability • Attention to Detail

PROJECTS

Healthcare AI - Dental Assessment Chatbot | [Node.js, Supabase, LLaMA 3.2B, Computer Vision, CSS] | GitHub

- Built an AI-driven dental assessment platform combining user inputs and intraoral images, reducing preliminary diagnosis time by ~60% using an LLM-powered chatbot (LLaMA 3.2B).
- Integrated appointment booking and patient history management via Supabase, supporting 3+ condition categories with sub-2s average response time.

Brain Tumor MRI Segmentation & Classification | [Python, Deep Learning, Attention U-Net, CNNs, Medical Imaging] | GitHub

- Built an end-to-end MRI analysis pipeline across 6,380 images spanning 4 tumor classes, performing 20+ hyperparameter tuning experiments to optimize loss functions.
- Achieved ~90% Dice score and ~95% classification accuracy through Attention ResUNet architecture and ensemble-based post-processing.

RAG-Based LLM Chatbot for Document QA | [Python, LLaMA 3.2, RAG, Qdrant, LangChain] | GitHub

- Built a RAG chatbot for document-based QA, integrating LLaMA 3.2 with Qdrant vector embeddings to handle 1,000+ document chunks with ~88% answer relevance.
- Designed an ingestion pipeline with semantic chunking and similarity search, reducing average query response time by 40% over keyword-based retrieval.

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional — Oracle (2025)
- Python Mega Course — Udemy (2023)
- Graph Theory Programming Bootcamp — AlgoUniversity (2023)

PUBLICATIONS

- Real-Time Web Chat Application with Live Performance Metrics* — IEEE (2024)
- Water Quality Monitoring System with Multiple Quality Testing Sensors* — IEEE (2024)
- Healthcare AI - Dental Assessment Chatbot* — IEEE (2025)